DataBase System

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- **Create a DML trigger that logs changes to a specific table when records are inserted, updated, or deleted.**

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```
CREATE TABLE emp_audit (
    emp_id NUMBER,
    action VARCHAR2(20),
    action_date DATE
);

CREATE OR REPLACE TRIGGER emp_audit_tr
AFTER INSERT OR UPDATE OR DELETE ON employees
FOR EACH ROW
BEGIN
    IF INSERTING THEN
        INSERT INTO emp_audit VALUES (:NEW.employee_id, 'INSERT', SYSDATE);
    ELSIF UPDATING THEN
        INSERT INTO emp_audit VALUES (:OLD.employee_id, 'UPDATE', SYSDATE);
    ELSIF DELETING THEN
        INSERT INTO emp_audit VALUES (:OLD.employee_id, 'DELETE', SYSDATE);
    END IF;
END;
/
```

```
Table EMP_AUDIT created.

Trigger EMP_AUDIT_TR compiled
```

- **Create a DML trigger that enforces a referential integrity constraint between two tables when inserting or updating records.**

```oracle
CREATE OR REPLACE TRIGGER check_dept_fk
BEFORE INSERT OR UPDATE ON employees
FOR EACH ROW
DECLARE
    v_count NUMBER;
BEGIN
    SELECT COUNT(*) INTO v_count
    FROM departments
    WHERE department_id = :NEW.department_id;

    IF v_count = 0 THEN
        RAISE_APPLICATION_ERROR(-20001, 'Invalid Department ID');
    END IF;
END;
/
```

```
Trigger CHECK_DEPT_FK compiled
```

- **Create a DML trigger that automatically updates a "last_modified" time stamp when a record is updated in a table.**

```oracle
ALTER TABLE employees ADD last_modified DATE;

CREATE OR REPLACE TRIGGER update_timestamp
BEFORE UPDATE ON employees
FOR EACH ROW
BEGIN
    :NEW.last_modified := SYSDATE;
END;
/
```

```
Table EMPLOYEES altered.

Trigger UPDATE_TIMESTAMP compiled
```

- **Create a DDL trigger that logs all schema changes (e.g., table creations, modifications, and drops) in a dedicated audit table.**

Worksheet Query Builder

```
CREATE TABLE ddl_audit (
     ddl_date DATE,
     user_name VARCHAR2(30),
     object_type VARCHAR2(30),
     object_name VARCHAR2(30),
     operation VARCHAR2(30)
);

CREATE OR REPLACE TRIGGER ddl_audit_tr
AFTER DDL ON SCHEMA
BEGIN
     INSERT INTO ddl_audit
     VALUES (SYSDATE,
             sys_context('USERENV','CURRENT_USER'),
             ora_dict_obj_type,
             ora_dict_obj_name,
             ora_sysevent);
END;
/
```

Script Output

Task completed in 0.038 seconds

```
Table DDL_AUDIT created.


Trigger DDL_AUDIT_TR compiled
```

- **Create a DDL trigger that prohibits any user from altering or dropping a specific critical table.**

Worksheet Query Builder

```
CREATE OR REPLACE TRIGGER protect_table
BEFORE DROP OR ALTER ON SCHEMA
BEGIN
    IF ora_dict_obj_name = 'EMPLOYEES' THEN
        RAISE_APPLICATION_ERROR(-20002, 'Cannot modify EMPLOYEES table');
    END IF;
END;
/
```

Script Output

Task completed in 0.035 seconds

```
Trigger PROTECT_TABLE compiled
```

- **Create a DDL trigger that prevents the creation of new tables with a specific naming pattern.**

Worksheet | Query Builder

```oracle_plsql
CREATE OR REPLACE TRIGGER prevent_table_creation
BEFORE CREATE ON SCHEMA
BEGIN
    IF ora_dict_obj_name LIKE 'TEST%' THEN
        RAISE_APPLICATION_ERROR(-20003, 'Table name not allowed');
    END IF;
END;
/
```

Script Output | Task completed in 0.036 seconds

```
Trigger PREVENT_TABLE_CREATION compiled
```

- **Create a system trigger that captures information when a user logs in.**

Worksheet | Query Builder

```oracle_plsql
CREATE TABLE login_audit (
    user_name VARCHAR2(30),
    login_time DATE
);

CREATE OR REPLACE TRIGGER login_tr
AFTER LOGON ON SCHEMA
BEGIN
    INSERT INTO login_audit
    VALUES (USER, SYSDATE);
END;
/
```

Script Output | Task completed in 0.035 seconds

```
Table LOGIN_AUDIT created.

Trigger LOGIN_TR compiled
```

- **Create a system trigger that sends an email notification to the DBA whenever a user with specific privileges logs in.**

```
CREATE OR REPLACE TRIGGER notify_dba
AFTER LOGON ON SCHEMA
BEGIN
    IF USER = 'HR' THEN
        DBMS_OUTPUT.PUT_LINE('Privileged user logged in!');
    END IF;
END;
/
```

Script Output

Task completed in 0.035 seconds

```
Trigger NOTIFY_DBA compiled
```

- **Create a system trigger that automatically sets the session time zone for all users upon login**

```
CREATE OR REPLACE TRIGGER set_timezone
AFTER LOGON ON SCHEMA
BEGIN
    EXECUTE IMMEDIATE 'ALTER SESSION SET TIME_ZONE = ''+05:00''';
END;
/
```

Script Output

Task completed in 0.046 seconds

```
Trigger SET_TIMEZONE compiled
```

- **Create an "Instead of" trigger for a view that allows inserting data into multiple tables when an insert operation is performed on the view.**

Worksheet | Query Builder

```
CREATE TABLE trainer (
    full_name VARCHAR2(30)
);

CREATE TABLE subject (
    subject_name VARCHAR2(30)
);


CREATE VIEW trainer_view AS
SELECT t.full_name, s.subject_name
FROM trainer t, subject s;



CREATE OR REPLACE TRIGGER insert_view_tr
INSTEAD OF INSERT ON trainer_view
FOR EACH ROW
BEGIN
    INSERT INTO trainer VALUES (:NEW.full_name);
    INSERT INTO subject VALUES (:NEW.subject_name);
END;
/
```

Script Output | Task completed in 0.048 seconds

```
Table TRAINER created.


Table SUBJECT created.


View TRAINER_VIEW created.


Trigger INSERT_VIEW_TR compiled
```

- **Create an "Instead of" trigger for a view that allows users to update records in a way that calculates and updates a computed column.**

Worksheet Query Builder

```
CREATE OR REPLACE TRIGGER update_view_tr
INSTEAD OF UPDATE ON trainer_view
FOR EACH ROW
BEGIN
    UPDATE trainer
    SET full_name = :NEW.full_name
    WHERE full_name = :OLD.full_name;

    UPDATE subject
    SET subject_name = :NEW.subject_name
    WHERE subject_name = :OLD.subject_name;
END;
/
```

Script Output

Task completed in 0.039 seconds

```
Trigger UPDATE_VIEW_TR compiled
```

- **Create an "Instead of" trigger for a view that allows users to insert records into multiple related tables through a single view.**

Worksheet Query Builder

```
CREATE OR REPLACE TRIGGER multi_insert_tr
INSTEAD OF INSERT ON trainer_view
FOR EACH ROW
BEGIN
    INSERT INTO trainer VALUES (:NEW.full_name);
    INSERT INTO subject VALUES (:NEW.subject_name);
END;
/
```

Script Output

Task completed in 0.043 seconds

```
Trigger MULTI_INSERT_TR compiled
```